

Friction Sensation in Redirected Walking Using a Rotating Handrail

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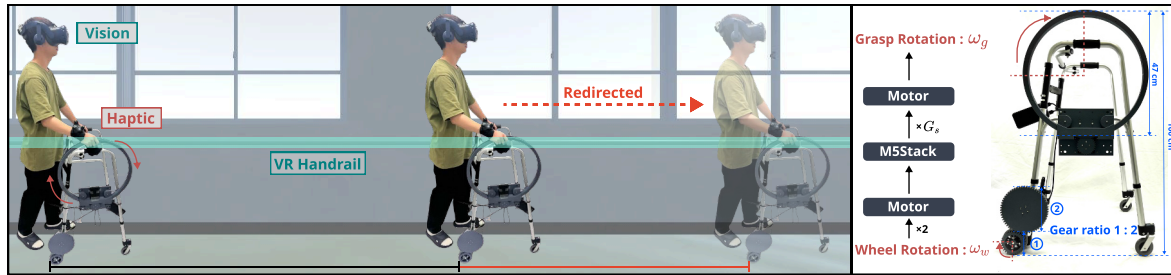


Figure 1: Our walker-type device provides a friction sensation with a rotating circular handrail. The slide gain can be adjusted dynamically, and the combination of visual and haptic feedback creates the illusion of translational movement for the user.

ABSTRACT

We propose a walker-type system designed to enhance Redirected Walking (RDW) within virtual reality (VR) by providing dynamic haptic feedback through a rotating circular handrail. Visual gain refers to a parameter that controls the amount of user movement within the virtual environment. Two user studies were conducted to investigate whether haptic feedback in RDW can expand the range of visual gain. In the first user study, we determined the thresholds of discomfort for slide gain (SG), defined as the ratio of the handrail's rotational amount to the amount of user moved. Results showed that slide gains up to $SG = 2.69$ could be presented without causing discomfort. The second user study examined the visual gain detection threshold in conditions with and without haptic feedback. This represents an approximately 23% increase in the visual gain threshold when the slide gain was set to 2.69, and a 21% increase when the slide gain was set to 1.00, highlighting the influence of haptic feedback. The results demonstrate that this device can enhance the user's perception of movement in VR by coupling visual and haptic cues.

Index Terms: Virtual Reality, Redirected Walking, Multimodal, Locomotion, Haptic device

1 INTRODUCTION

Free walking in a vast virtual environment requires an equally vast real environment, making it difficult to simulate natural walking [4]. One solution to this problem is a technique called Redirected Walking (RDW), which subtly alters the distance and di-

rection of movement by manipulating the visual information presented to the user [5]. The primary advantage of RDW is that it allows users to experience accurate proprioceptive, kinesthetic, and vestibular stimuli in the physical world. In general, RDW manipulates a user's movement by adjusting gains, defined as the ratio of virtual movement or rotation to physical movement or rotation. These gains modify the user's perceived movement in the virtual environment, allowing them to experience greater distances and rotations compared to their actual physical movement. However, it has been shown that when translational gain exceeds 1.22, users become aware of discomfort, and immersion is reduced [6]. The impact of these translational gains is weaker compared to rotational and curvature gains, making the extension of the usable space small and impractical.

To improve the effectiveness of RDW, some research explores the impacts of multisensory cues on the rotational gain. Matsumoto et al. proposed the Unlimited Corridor [3], which allows users to walk in a large virtual environment without interruption by presenting images of walking straight ahead while their hands touch a cylindrical wall. However, no studies have investigated RDW on the translational gain by combining visual and haptic feedback.

Therefore, we focused on the friction sensation along a handrail as continuous haptic feedback during translational motion. Walking while gripping a handrail is a common everyday action, and it allows for providing haptic feedback without compromising the user experience. We hypothesized that dynamically adjusting the friction feedback could provide a haptic experience that corresponds to the perceived movement distance. This study develops a device that can present frictional sensations while walking with a handrail, and investigates the manipulation of straight-line movement through the interaction of visual and haptic cues.

2 DESIGN AND IMPLEMENTATIONS

The device consists of two main components: the grip and the wheel, and incorporates two motors and an M5Stack microcontroller. The motors employed are outer rotor motors (KM-1U) with built-in magnetic rotary encoders. These motors can achieve a maximum rotational speed of 720rpm and deliver a maximum torque of 0.1Nm. The M5Stack microcontroller is used to control the rota-

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tional speed of these motors.

In the wheel assembly, the rotation of the walker's tires is tracked using gears attached to the tires. The rotational data from the gears is transmitted to the motor connected to the wheel, which in turn relays the data to the grip component.

The circular grip, fabricated using a 3D printer, has a diameter of 47cm and is positioned at a height of 100cm from the ground. It rotates via a gear-driven mechanism powered by the motors, simulating a frictional sensation along the handrail that replicates the tactile experience of friction.

This device provides a frictional sensation with adjustable rotational levels, calibrated based on the user's movement. The slide gain represents a parameter that modulates the intensity of the haptic frictional sensation experienced while gripping the handrail.

A slide gain ranging from 0.10 to 4.00 is applied to the rotational data from the gear attached to the wheel. This adjusted rotational data is then transmitted via the M5Stack to the gear mechanism within the grip. By applying the slide gain to the angular velocity of the gear connected to the tire, the angular velocity of the grip gear is precisely controlled, ensuring a realistic and responsive tactile experience.

The formula for calculating the rotation amount of the grip section is as follows:

$$\omega_g = 2 \cdot \omega_w \cdot G_s \quad (1)$$

where ω_g represents the rotational amount of the motor in the grip section, and ω_w represents the rotational amount of the motor in the wheel section. In the wheel section, the gear attached to the tire and the gear attached to the rotation measurement motor have a ratio of 1 : 2. Hence, the factor of 2 is included in the formula. The slide gain G_s , which adjusts the rotational amount, is controlled through serial communication with Unity, ensuring appropriate values are provided according to the experimental conditions.

3 USER STUDY 1: COMFORTABLE SLIDE GAIN

First user study aims to investigate the range of slide gain values that can be presented without causing discomfort when using this system. A total of 16 participants (9 males and 7 females) took part in the user study, with an average age of 23.4 years ($SD = 1.32$). They wore a HTC VIVE Pro HMD and standing while gripping the device with their right hand. Participants walked straight for five meters while gripping the device and interacting with the virtual handrail in VR. In the staircase task, the adjustable range of slide gain for the high condition was set from 1.00 to 4.00, while for the low condition, it was set from 0.25 to 1.00. In this study, the visual Gain was fixed at 1.00. After each walk, they completed a questionnaire to assess the match between visual and haptic sensations. For this study, we employed the staircase method to evaluate discomfort. Ethical approval for this study was obtained from our Institutional Review Board with IRB number 2023-I-32.

3.1 Analysis and Results

Using the staircase method, the threshold values for $SG_H = 2.69 \pm 1.30$ and $SG_L = 0.54 \pm 0.29$ were determined. These results indicate that humans can tolerate a relatively fast haptic sensation of friction on the handrail without feeling discomfort.

4 USER STUDY 2: UNDETECTABLE VISUAL GAIN

Second user study investigated the maximum undetectable visual gain under three conditions using the same environment as User Study 1: when the slide gain was set to the maximum comfortable value obtained in User Study 1 ($SG_H = 2.69$), when the slide gain matched the walking speed ($SG_L = 1.00$), and when no handrail was used (SG_N). A total of 24 participants (12 males and 12 females) took part in the user study, with an average age of 23.3

years ($SD = 2.24$). In the adaptive staircase method task, participants in the SG_H and SG_S conditions walked five meters while holding the device. After each walk, they answered two questionnaires to assess whether they noticed the manipulation of the visual feedback. Ethical approval for this study was obtained from our Institutional Review Board with IRB number 2023-I-32.

4.1 Analysis and Results

The visual gain detection threshold was 2.789 ± 1.110 for the SG_H condition, 2.310 ± 0.855 for the SG_S condition, and 2.256 ± 1.011 for the SG_N condition. The results showed between the SG_H and SG_S conditions ($p < 0.003$), between the SG_H and SG_N conditions ($p < 0.001$). Walking speeds were also compared for each condition at $VG = 1.00$. The results showed between the SG_H and SG_N conditions ($p < 0.01$), between the SG_S and SG_N conditions ($p < 0.01$). We performed a Bonferroni correction to account for multiple comparisons, confirming significant differences between SG_H and SG_N , and SG_S and SG_N .

5 CONCLUSION AND FUTURE WORK

We developed and evaluated a new haptic feedback method for translational RDW. Our walker-type device allows users to walk while gripping a handrail, with the ability to dynamically adjust the friction sensation (slide gain) of the rotating circular handrail. In the subsequent investigation of the visual gain detection threshold using this device, when the $SG = 2.69$, the visual gain reached 2.78 ± 1.11 , suggesting that visual manipulation could be increased by approximately 23% compared to conditions without haptic feedback. The results demonstrate that it can enhance the user's perception of movement in VR by coupling visual and haptic cues. One of key limitations of this device is the limited degrees of freedom in the walking direction. The current device is limited to straight-line movement, making lateral movement impossible. This restriction reduces the device's versatility, and increasing its mobility could significantly contribute to advancements in the RDW field. This is exemplified by studies showing an enhancement in the effects of RDW by combining translational and curvature manipulation [1]. For example, Matsumoto et al. [2] presented handrail feedback combined with curvature RDW. Therefore, incorporating this device with these methods could further enhance RDW effectiveness.

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